

Sample Data

Sample Table: **Employees**

EmpID	EmpName	Department	City	Salary	HireDate
101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
102	Priya Sharma	HR	Mumbai	62000	2019-09-25
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
104	Neha Patel	Sales	Delhi	48000	2022-01-14
105	Karan Joshi	Marketing	Pune	45000	2018-07-22
106	Divya Nair	IT	Chennai	81000	2019-12-11
107	Raj Kumar	HR	Delhi	60000	2020-05-28
108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
110	Anjali Das	Sales	Kolkata	51000	2023-01-15

1. Show employees working in either the 'IT' or 'HR' departments.

Answer:

To show employees working in either the 'IT' or 'HR' departments.

select *

from Employees

where Department = 'IT' or Department = 'HR';

The screenshot shows a database management tool interface. On the left, there is a 'SCHEMAS' panel with a tree view containing various database objects like 'account1', 'account2', 'account3', 'brilliantchoc', 'classes', 'company_db', 'db', 'employees', 'Tables', 'Views', 'Stored Pr', 'Functions', and 'join demo'. The 'employees' table is selected. The main area displays a SQL query:

```

22  (109, 'Arjun Reddy', 'IT', 'Hyderabad', 70000, '2022-02-18'),
23  (110, 'Anjali Das', 'Sales', 'Kolkata', 51000, '2023-01-15');
24
25
26
27  select *
28  from Employees
29  where Department = 'IT' or Department = 'HR';

```

Below the query, the 'Result Grid' shows the following data:

EmpID	EmpName	Department	City	Salary	HireDate
102	Priya Sharma	HR	Mumbai	62000	2019-09-25
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
106	Divya Nair	IT	Chennai	81000	2019-12-11
107	Raj Kumar	HR	Delhi	60000	2020-05-28
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
*	NULL	NULL	NULL	NULL	NULL

The interface also includes a 'Filter Rows' section and a 'No object selected' message at the bottom left.

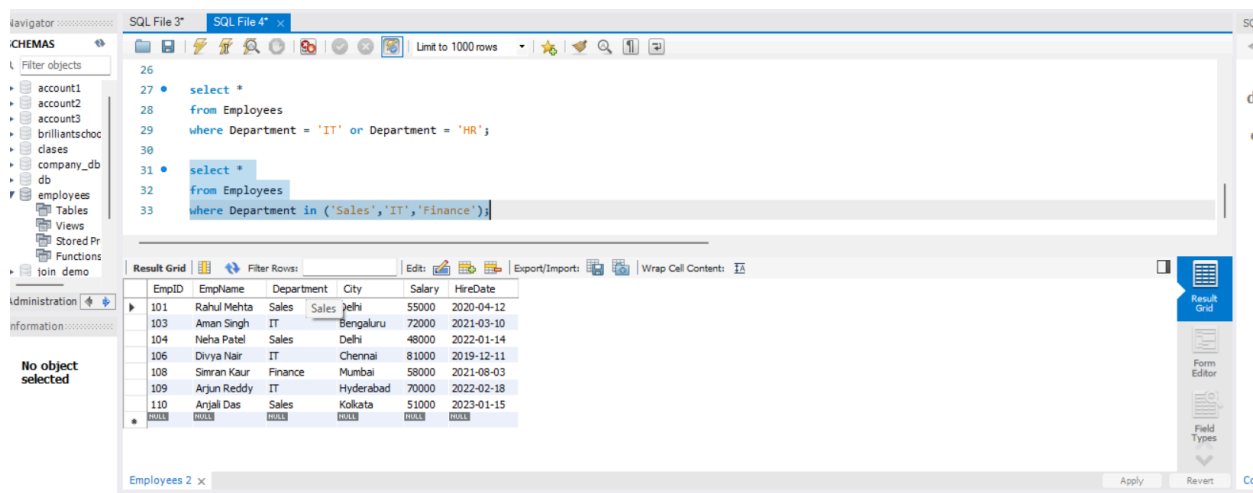
This Query checks the department column and shows only those employees who belongs to IT or HR.

2. Retrieve employees whose department is in 'Sales', 'IT', or 'Finance'.

Answer:

Retrive employees whose department is in 'Sales','IT', or 'Finance'.

```
select *
from Employees
where Department in ('Sales','IT','Finance');
```



The screenshot shows a SQL IDE interface with a query editor and a result grid. The query editor contains the following SQL code:

```
26
27 • select *
28   from Employees
29   where Department = 'IT' or Department = 'HR';
30
31 • select *
32   from Employees
33   where Department in ('Sales','IT','Finance');
```

The result grid displays the following data:

EmpID	EmpName	Department	City	Salary	HireDate
101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
104	Neha Patel	Sales	Delhi	48000	2022-01-14
106	Divya Nair	IT	Chennai	81000	2019-12-11
108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
110	Anjali Das	Sales	Kolkata	51000	2023-01-15

Here we IN instead of multiple OR conditions. Using IN condition it makes shorter and cleaner.

3. Display employees whose salary is between ₹50,000 and ₹70,000.

Answer:

Employees salaries between 50,000 and 70,000.

```
select *
from Employees
where Salary between 50000 and 70000;
```

The screenshot shows a Google Colab environment with a file explorer on the left displaying a database structure. The main area contains a SQL query in a code cell:

```

34
35 • select *
36 from Employees
37 where Salary between 50000 and 70000;

```

Below the code cell is a 'Result Grid' showing the output of the query. The grid has columns: EmpID, EmpName, Department, City, Salary, and HireDate. The results are as follows:

EmpID	EmpName	Department	City	Salary	HireDate
101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
102	Priya Sharma	HR	Mumbai	62000	2019-09-25
107	Raj Kumar	HR	Delhi	60000	2020-05-28
108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
110	Anjali Das	Sales	Kolkata	51000	2023-01-15
NULL	NULL	NULL	NULL	NULL	NULL

This will give employees whose salary lies in between 50000 and 70000.

4. List employees whose names start with the letter 'A'.

Answer:

To list employees whose names start with the letter 'A'.

```

select *
from Employees
where EmpName like 'A%';

```

The screenshot shows a Google Colab environment with a file explorer on the left displaying a database structure. The main area contains a SQL query in a code cell:

```

39 • select *
40 from Employees
41 where EmpName like 'A%';

```

Below the code cell is a 'Result Grid' showing the output of the query. The grid has columns: EmpID, EmpName, Department, City, Salary, and HireDate. The results are as follows:

EmpID	EmpName	Department	City	Salary	HireDate
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
110	Anjali Das	Sales	Kolkata	51000	2023-01-15
NULL	NULL	NULL	NULL	NULL	NULL

Here % refers to anything that can come after 'A', so it filters names starting with A.

5. Find employees whose names contain the substring 'an'.

Answer:

To find employees whose names contain the substring 'an'.

```
select *
from Employees
where EmpName like '%an%';
```

The screenshot shows a Jupyter Notebook interface. On the left, there is a file explorer showing a directory structure with folders like 'classes', 'company_db', 'db', and 'employees'. The 'employees' folder is expanded, showing 'Tables', 'Views', 'Stored Pr', and 'Functions'. Below this, there are buttons for 'Administration' and 'Information'. A message 'No object selected' is displayed. The main area shows a SQL query in a code cell:

```
42
43 • select *
44 from Employees
45 where EmpName like '%an%';
```

Below the code cell is a 'Result Grid' showing the results of the query. The grid has columns: EmpID, EmpName, Department, City, Salary, and HireDate. The results are as follows:

EmpID	EmpName	Department	City	Salary	HireDate
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
105	Karan Joshi	Marketing	Pune	45000	2018-07-22
108	Simran Kaur	Finance	Marketing	58000	2021-08-03
110	Anjali Das	Sales	Kolkata	51000	2023-01-15
NULL	NULL	NULL	NULL	NULL	NULL

This checks if "an" is present anywhere in the name.

6. Show employees who are from 'Delhi' or 'Mumbai' and earn more than ₹55,000.

Answer:

To show employees who are from 'Delhi' or 'Mumbai' and earn more than 50000.

```
select *
from employees
where (City = 'Delhi' or City = 'Mumbai')
and Salary >55000;
```

The screenshot shows a Jupyter Notebook interface. On the left, there is a file explorer showing a directory structure with folders like 'brilliantshoc', 'classes', 'company_db', 'db', and 'employees'. The 'employees' folder is expanded, showing 'Tables', 'Views', 'Stored Pr', and 'Functions'. Below this, there are buttons for 'Administration' and 'Information'. A message 'No object selected' is displayed. The main area shows a SQL query in a code cell:

```
46
47 • select *
48 from employees
49 where (City = 'Delhi' or City = 'Mumbai')
50 and Salary >55000;
```

Below the code cell is a 'Result Grid' showing the results of the query. The grid has columns: EmpID, EmpName, Department, City, Salary, and HireDate. The results are as follows:

EmpID	EmpName	Department	City	Salary	HireDate
102	Priya She	HR	Delhi	60000	2020-05-28
107	Raj Kumar	HR	Delhi	60000	2020-05-28
108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
NULL	NULL	NULL	NULL	NULL	NULL

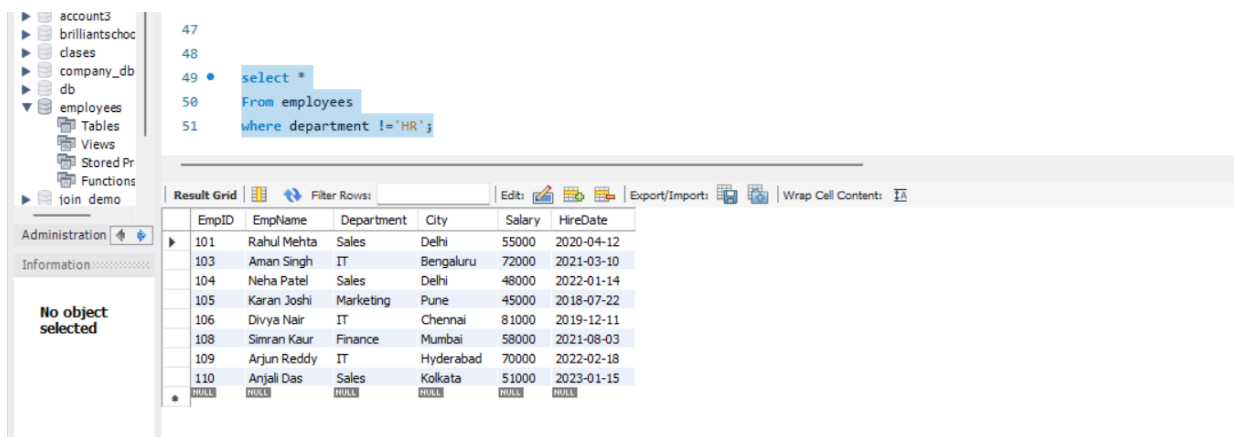
First it filters cities, then applies salary condition.

7. Display all employees except those from the 'HR' department.

Answer:

To display all employees except those from the 'HR' department.

```
select *
From employees
where department != 'HR';
```



The screenshot shows a database interface with a left sidebar containing a tree view of database objects (account3, brilliantschoc, classes, company_db, db, employees, Tables, Views, Stored Pr, Functions, join demo). The main area displays a SQL query:

```
47
48
49 • select *
50 from employees
51 where department != 'HR';
```

Below the query is a "Result Grid" with the following data:

EmpID	EmpName	Department	City	Salary	HireDate
101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
104	Neha Patel	Sales	Delhi	48000	2022-01-14
105	Karan Joshi	Marketing	Pune	45000	2018-07-22
106	Divya Nair	IT	Chennai	81000	2019-12-11
108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
110	Anjali Das	Sales	Kolkata	51000	2023-01-15
*	NULL	NULL	NULL	NULL	NULL

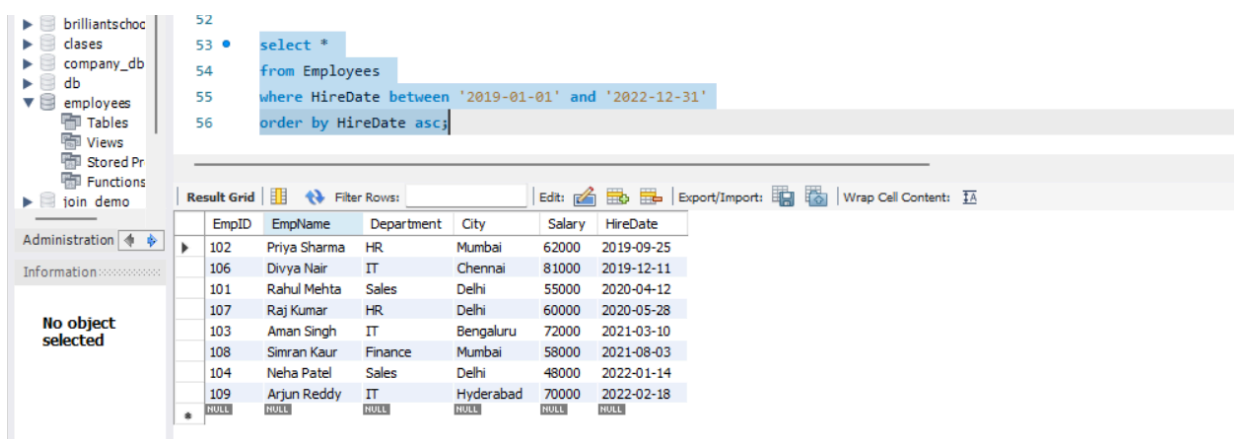
These removes HR employees and shows the rest.

8. Get all employees hired between 2019 and 2022, ordered by HireDate (oldest first).

Answer:

To get all employees hired between 2019 and 2022, ordered by Hiredate (oldest first).

```
select *
from Employees
where HireDate between '2019-01-01' and '2022-12-31'
order by HireDate asc;
```



The screenshot shows the same database interface with a different SQL query:

```
52
53 • select *
54 from Employees
55 where HireDate between '2019-01-01' and '2022-12-31'
56 order by HireDate asc;
```

The "Result Grid" shows the following data, ordered by HireDate:

EmpID	EmpName	Department	City	Salary	HireDate
102	Priya Sharma	HR	Mumbai	62000	2019-09-25
106	Divya Nair	IT	Chennai	81000	2019-12-11
101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
107	Raj Kumar	HR	Delhi	60000	2020-05-28
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
104	Neha Patel	Sales	Delhi	48000	2022-01-14
109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
*	NULL	NULL	NULL	NULL	NULL

This filters based on date range and sorts from oldest to newest.

